

FIT5197 2026 S1 Final Assessment

SPECIAL NOTE: Please refer to the [assessment page \(Clayton\)](#), or [\(Malaysia\)](#) for rules, general guidelines and marking rubrics of the assessment (the marking rubric for the kaggle competition part will be released near the deadline – around 5 working days – in the same page). Failure to comply with the provided information will result in a deduction of mark (e.g., late penalties) or breach of academic integrity.

PLEASE ENSURE YOU READ AND COMPLETE ALL REQUIRED INFORMATION BELOW BEFORE STARTING YOUR ASSIGNMENT, AS FAILURE TO DO SO WILL RESULT IN A MARK OF ZERO.

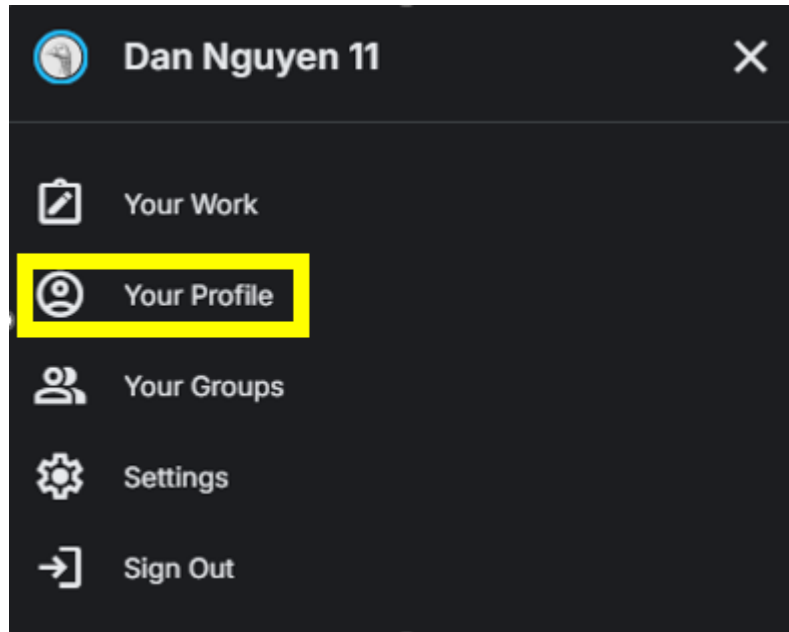
YOUR NAME: Keshu Zhang

STUDENT ID: 36436763

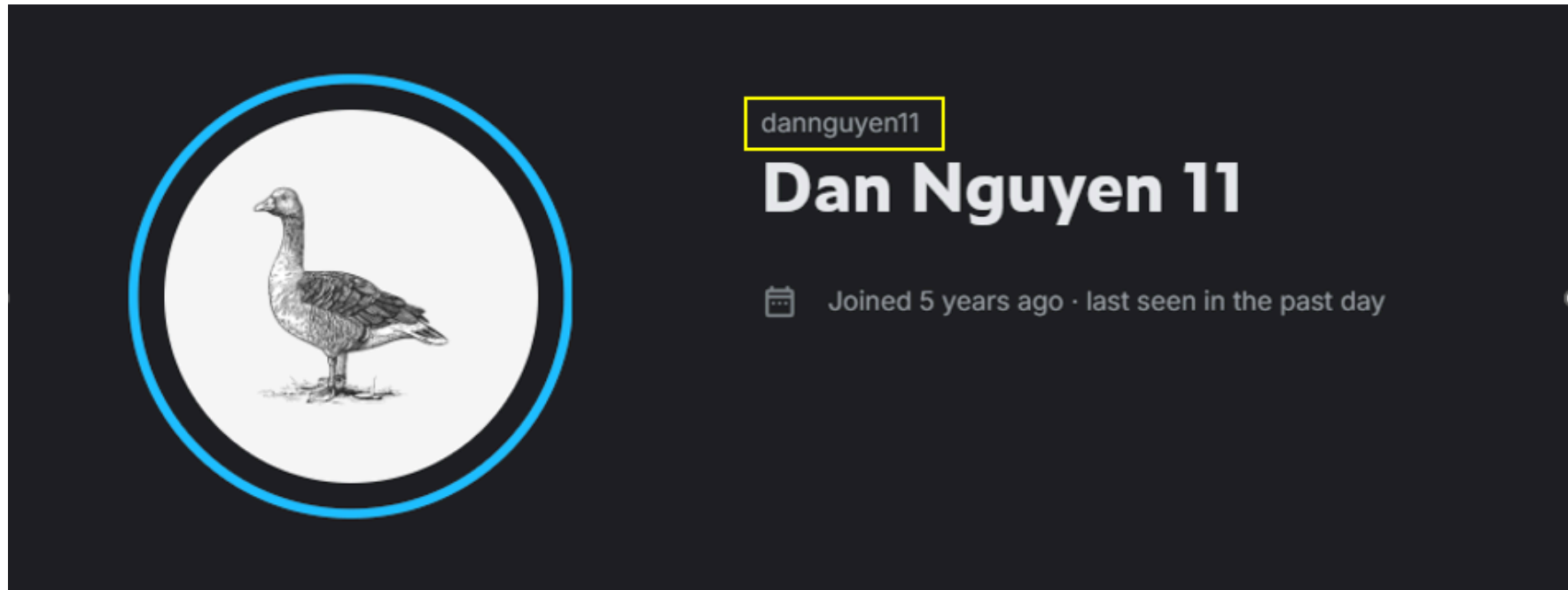
KAGGLE NAME (required for marking – see Part 1, Question 5 or Part 2): zksdawn4pm

Important: Your Kaggle name must be identical (including capitalization, spacing, and special characters) across both Regression and Classification contests. Marks will be deducted if your Kaggle name is inconsistent or missing. No exceptions will be made!

You can find your Kaggle name by clicking on your profile icon at the top-right corner of the Kaggle website and go to your profile. For example, here is a screenshot of where to find it:



After that, the Kaggle name can be found similar to the one in the picture:



Please note that the name to be recorded is not "Dan Nguyen 11" but "dannguyen11". Marks will be deducted if you record the incorrect names.

Please also enter your details in this [google form](#).

Part 1 Regression (50 Marks)

A few thousand people were questioned in a life and wellbeing survey to build a model to predict happiness of an individual. You need to build regression models to optimally predict the variable in the survey dataset called 'happiness' based on any, or all, of the other survey question responses.

You have been provided with two datasets, `regression_train.csv` and `regression_test.csv`. Using these datasets, you hope to build a model that can predict happiness level using the other variables. `regression_train.csv` comes with the ground-truth target label (i.e. happiness level) whereas `regression_test.csv` comes with independent variables (input information) only.

On the order of around 70 survey questions have been converted into predictor variables that can be used to predict happiness. We do not list all the predictor names here, but their names given in the data header can clearly be linked to the survey questions. e.g., the predictor variable 'iDontFeelParticularlyPleasedWithTheWayIAm' corresponds to the survey question 'I don't feel particularly pleased with the way I am.'

PLEASE NOTE THAT THE USE OF LIBRARIES ARE PROHIBITED IN THESE QUESTIONS UNLESS STATED OTHERWISE, ANSWERS USING LIBRARIES WILL RECEIVE 0 MARKS

Question 1 (NO LIBRARIES ALLOWED) (4 Mark)

Please load the `regression_train.csv` and fit a **multiple linear regression model** with 'happiness' being the target variable. According to the summary table, which predictors do you think are possibly associated with the target variable (use the significance level of 0.01), and which are the **Top 5** strongest predictors? Please write an R script to automatically fetch and print this information.

NOTE: Manually doing the above tasks will result in 0 marks.

```
In [16]: #get file path for local or Kaggle environment
get_data_path_reg <- function(file_name) {
  if (file.exists(file_name)) {
    return(file_name)
  }

  kaggle_path <- file.path(
    "/kaggle/input/datasets/zksdawn4pm/regression-dataset",
    file_name
  )

  if (file.exists(kaggle_path)) {
    return(kaggle_path)
  }

  stop(paste("Cannot find file:", file_name))
}
```

```
In [17]: # ANSWER BLOCK
```

```
regression_train <- read.csv(get_data_path_reg("regression_train.csv"))
regression_test  <- read.csv(get_data_path_reg("regression_test.csv"))

full_model<- lm(happiness ~ ., data = regression_train)

sum_full_model<-summary(full_model)
sum_full_model

coefficients_table <- sum_full_model$coefficients
predictors_table <- coefficients_table[rownames(coefficients_table)!="(Intercept)", , drop = FALSE]
significant_predictors <- predictors_table[predictors_table[, "Pr(>|t|)"] < 0.01, , drop = FALSE]

print(rownames(significant_predictors))

top5_strongest <- head(predictors_table[order(abs(predictors_table[, "t value"]), decreasing = TRUE), , drop = FALSE], 5)

cat("Top 5 strongest predictors are :",rownames(top5_strongest))
top5_strongest
```

Call:

```
lm(formula = happiness ~ ., data = regression_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-27.9439	-4.2430	0.4693	4.3861	21.7657

Coefficients:

	Estimate
(Intercept)	-21.88359
genderMale	-0.30236
genderOther	-1.00174
income10k - 15k	6.28585
income120k - 150k	11.47689
income150k - 200k	11.45754
income15k - 20k	15.40746
income200k above	20.03062
income20k - 50k	20.81995
income50k - 80k	28.70233
income80k - 120k	37.28064
whatIsYourHeightExpressItAsANumberInMetresM150 - 155	1.39209
whatIsYourHeightExpressItAsANumberInMetresM155 - 160	2.31478
whatIsYourHeightExpressItAsANumberInMetresM160 - 165	2.68535
whatIsYourHeightExpressItAsANumberInMetresM165 - 170	2.83607
whatIsYourHeightExpressItAsANumberInMetresM170 - 175	4.26762
whatIsYourHeightExpressItAsANumberInMetresM175 - 180	5.65960
whatIsYourHeightExpressItAsANumberInMetresM180 - 185	6.94023
whatIsYourHeightExpressItAsANumberInMetresM185 - 190	6.28022
whatIsYourHeightExpressItAsANumberInMetresM190 above	3.40392
doYouFeelASenseOfPurposeAndMeaningInYourLife104Yes	3.59908
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey complement each other	0.01440
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey conflict	1.56590
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsNever	-1.48209
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsRarely	-1.20871
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsSometimes	0.04914
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have both	1.40125
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have family	1.39367
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have friends	2.11356
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVAt least once a week	0.25673
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVMore than three times a week	0.01004
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVNo	1.56547

howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVRarely	1.19699
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferNever	-0.35247
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferRarely	-1.72271
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferSometimes	-2.01676
doYouFeelASenseOfPurposeAndMeaningInYourLife105Yes	0.76883
alwaysStressed	-1.34118
alwaysAccountableAndResponsibleForYourActions	0.65927
alwaysCalm	0.27971
myBodyIsHypermobileAndLovesToMove	0.19965
alwaysHaveFun	0.49673
alwaysSerious	-0.49491
alwaysDepressed	-0.65385
alwaysLoveAndCareForYourself	-0.12965
extremelyGoodAbilityToSense	-0.17572
alwaysHaveDigestiveProblems	0.18605
iAmIntenselyInterestedInOtherPeople	0.13554
iRarelyWakeUpFeelingRested	-0.40786
iFindMostThingsAmusing	0.54922
iAmAlwaysCommittedAndInvolved	-0.89999
iDoNotThinkThatTheWorldIsAGoodPlace	-0.61153
iAmWellSatisfiedAboutEverythingInMyLife	0.38435
iFindBeautyInSomeThings	0.28395
iAlwaysHaveACheerfulEffectOnOthers	0.86138
iFeelThatIAmNotEspeciallyInControlOfMyLife	-0.37252
iUsuallyHaveAGoodInfluenceOnEvents	0.65674
iDontHaveFunWithOtherPeople	-0.24931
alwaysEngageInPreparingAndUsingYourSkillsAndTalentsInOrderToGai	-0.10213
alwaysMakingProgress	-0.42485
extremelyGoodCommunicator	-0.39350
iDontFeelParticularlyPleasedWithTheWayIAm	-0.33182
iFeelThatLifeIsVeryRewarding	-0.27792
iHaveVeryWarmFeelingsTowardsAlmostEveryone	0.06943
iAmNotParticularlyOptimisticAboutTheFuture	-0.07469
lifeIsGood	0.45618
iLaughALot	0.80375
iDontThinkILookAttractive	0.45982
	Std. Error
(Intercept)	3.52855
genderMale	0.88129
genderOther	2.37254
income10k - 15k	1.24263

income120k – 150k	1.23683
income150k – 200k	1.24011
income15k – 20k	1.28486
income200k above	1.38264
income20k – 50k	1.58755
income50k – 80k	1.74201
income80k – 120k	1.66005
whatIsYourHeightExpressItAsANumberInMetresM150 – 155	2.32747
whatIsYourHeightExpressItAsANumberInMetresM155 – 160	2.13866
whatIsYourHeightExpressItAsANumberInMetresM160 – 165	2.11694
whatIsYourHeightExpressItAsANumberInMetresM165 – 170	2.11575
whatIsYourHeightExpressItAsANumberInMetresM170 – 175	2.15960
whatIsYourHeightExpressItAsANumberInMetresM175 – 180	2.20126
whatIsYourHeightExpressItAsANumberInMetresM180 – 185	2.38009
whatIsYourHeightExpressItAsANumberInMetresM185 – 190	2.98616
whatIsYourHeightExpressItAsANumberInMetresM190 above	4.38080
doYouFeelASenseOfPurposeAndMeaningInYourLife104Yes	1.95918
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey complement each other	0.81202
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey conflict	1.15356
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsNever	1.33228
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsRarely	1.20708
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsSometimes	1.19736
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have both	1.19698
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have family	1.34259
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have friends	2.11928
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVAt least once a week	0.97803
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVMore than three times a week	1.05138
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVNo	1.20553
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVRarely	1.36361
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferNever	0.99812
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferRarely	1.00573
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferSometimes	0.99350
doYouFeelASenseOfPurposeAndMeaningInYourLife105Yes	0.96165
alwaysStressed	0.34541
alwaysAccountableAndResponsibleForYourActions	0.43870
alwaysCalm	0.32603
myBodyIsHypermobileAndLovesToMove	0.30679
alwaysHaveFun	0.35477
alwaysSerious	0.31807
alwaysDepressed	0.38610
alwaysLoveAndCareForYourself	0.37296

extremelyGoodAbilityToSense	0.37718
alwaysHaveDigestiveProblems	0.33773
iAmIntenselyInterestedInOtherPeople	0.32171
iRarelyWakeUpFeelingRested	0.30388
iFindMostThingsAmusing	0.33887
iAmAlwaysCommittedAndInvolved	0.36946
iDoNotThinkThatTheWorldIsAGoodPlace	0.33844
iAmWellSatisfiedAboutEverythingInMyLife	0.36342
iFindBeautyInSomeThings	0.52685
iAlwaysHaveACheerfulEffectOnOthers	0.36405
iFeelThatIAmNotEspeciallyInControlOfMyLife	0.33673
iUsuallyHaveAGoodInfluenceOnEvents	0.36401
iDontHaveFunWithOtherPeople	0.36954
alwaysEngageInPreparingAndUsingYourSkillsAndTalentsInOrderToGain	0.35679
alwaysMakingProgress	0.38959
extremelyGoodCommunicator	0.34051
iDontFeelParticularlyPleasedWithTheWayIAm	0.32534
iFeelThatLifeIsVeryRewarding	0.38021
iHaveVeryWarmFeelingsTowardsAlmostEveryone	0.33195
iAmNotParticularlyOptimisticAboutTheFuture	0.32415
lifeIsGood	0.45211
iLaughALot	0.37074
iDontThinkILookAttractive	0.32252
t value	
(Intercept)	-6.202
genderMale	-0.343
genderOther	-0.422
income10k - 15k	5.059
income120k - 150k	9.279
income150k - 200k	9.239
income15k - 20k	11.992
income200k above	14.487
income20k - 50k	13.115
income50k - 80k	16.477
income80k - 120k	22.458
whatIsYourHeightExpressItAsANumberInMetresM150 - 155	0.598
whatIsYourHeightExpressItAsANumberInMetresM155 - 160	1.082
whatIsYourHeightExpressItAsANumberInMetresM160 - 165	1.269
whatIsYourHeightExpressItAsANumberInMetresM165 - 170	1.340
whatIsYourHeightExpressItAsANumberInMetresM170 - 175	1.976
whatIsYourHeightExpressItAsANumberInMetresM175 - 180	2.571

whatIsYourHeightExpressItAsANumberInMetresM180 - 185	2.916
whatIsYourHeightExpressItAsANumberInMetresM185 - 190	2.103
whatIsYourHeightExpressItAsANumberInMetresM190 above	0.777
doYouFeelASenseOfPurposeAndMeaningInYourLife104Yes	1.837
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey complement each other	0.018
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey conflict	1.357
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsNever	-1.112
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsRarely	-1.001
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsSometimes	0.041
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have both	1.171
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have family	1.038
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have friends	0.997
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVAt least once a week	0.263
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVMore than three times a week	0.010
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVNo	1.299
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVRarely	0.878
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferNever	-0.353
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferRarely	-1.713
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferSometimes	-2.030
doYouFeelASenseOfPurposeAndMeaningInYourLife105Yes	0.799
alwaysStressed	-3.883
alwaysAccountableAndResponsibleForYourActions	1.503
alwaysCalm	0.858
myBodyIsHypermobileAndLovesToMove	0.651
alwaysHaveFun	1.400
alwaysSerious	-1.556
alwaysDepressed	-1.693
alwaysLoveAndCareForYourself	-0.348
extremelyGoodAbilityToSense	-0.466
alwaysHaveDigestiveProblems	0.551
iAmIntenselyInterestedInOtherPeople	0.421
iRarelyWakeUpFeelingRested	-1.342
iFindMostThingsAmusing	1.621
iAmAlwaysCommittedAndInvolved	-2.436
iDoNotThinkThatTheWorldIsAGoodPlace	-1.807
iAmWellSatisfiedAboutEverythingInMyLife	1.058
iFindBeautyInSomeThings	0.539
iAlwaysHaveACheerfulEffectOnOthers	2.366
iFeelThatIAmNotEspeciallyInControlOfMyLife	-1.106
iUsuallyHaveAGoodInfluenceOnEvents	1.804
iDontHaveFunWithOtherPeople	-0.675

alwaysEngageInPreparingAndUsingYourSkillsAndTalentsInOrderToGai	-0.286
alwaysMakingProgress	-1.091
extremelyGoodCommunicator	-1.156
iDontFeelParticularlyPleasedWithTheWayIAm	-1.020
iFeelThatLifeIsVeryRewarding	-0.731
iHaveVeryWarmFeelingsTowardsAlmostEveryone	0.209
iAmNotParticularlyOptimisticAboutTheFuture	-0.230
lifeIsGood	1.009
iLaughALot	2.168
iDontThinkILookAttractive	1.426
	Pr(> t)
(Intercept)	1.31e-09
genderMale	0.731697
genderOther	0.673072
income10k - 15k	6.26e-07
income120k - 150k	< 2e-16
income150k - 200k	< 2e-16
income15k - 20k	< 2e-16
income200k above	< 2e-16
income20k - 50k	< 2e-16
income50k - 80k	< 2e-16
income80k - 120k	< 2e-16
whatIsYourHeightExpressItAsANumberInMetresM150 - 155	0.550079
whatIsYourHeightExpressItAsANumberInMetresM155 - 160	0.279701
whatIsYourHeightExpressItAsANumberInMetresM160 - 165	0.205300
whatIsYourHeightExpressItAsANumberInMetresM165 - 170	0.180802
whatIsYourHeightExpressItAsANumberInMetresM170 - 175	0.048778
whatIsYourHeightExpressItAsANumberInMetresM175 - 180	0.010472
whatIsYourHeightExpressItAsANumberInMetresM180 - 185	0.003731
whatIsYourHeightExpressItAsANumberInMetresM185 - 190	0.036034
whatIsYourHeightExpressItAsANumberInMetresM190 above	0.437579
doYouFeelASenseOfPurposeAndMeaningInYourLife104Yes	0.066891
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey complement each other	0.985860
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey conflict	0.175347
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsNever	0.266565
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsRarely	0.317221
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsSometimes	0.967284
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have both	0.242382
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have family	0.299827
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have friends	0.319176
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVAt least once a week	0.793061

howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVMore than three times a week	0.992388
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVNo	0.194782
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVRarely	0.380533
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferNever	0.724162
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferRarely	0.087447
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferSometimes	0.042975
doYouFeelASenseOfPurposeAndMeaningInYourLife105Yes	0.424447
alwaysStressed	0.000119
alwaysAccountableAndResponsibleForYourActions	0.133628
alwaysCalm	0.391410
myBodyIsHypermobilityAndLovesToMove	0.515542
alwaysHaveFun	0.162184
alwaysSerious	0.120446
alwaysDepressed	0.091081
alwaysLoveAndCareForYourself	0.728293
extremelyGoodAbilityToSense	0.641542
alwaysHaveDigestiveProblems	0.581999
iAmIntenselyInterestedInOtherPeople	0.673740
iRarelyWakeUpFeelingRested	0.180240
iFindMostThingsAmusing	0.105800
iAmAlwaysCommittedAndInvolved	0.015256
iDoNotThinkThatTheWorldIsAGoodPlace	0.071473
iAmWellSatisfiedAboutEverythingInMyLife	0.290828
iFindBeautyInSomeThings	0.590198
iAlwaysHaveACheerfulEffectOnOthers	0.018418
iFeelThatIAmNotEspeciallyInControlOfMyLife	0.269208
iUsuallyHaveAGoodInfluenceOnEvents	0.071896
iDontHaveFunWithOtherPeople	0.500272
alwaysEngageInPreparingAndUsingYourSkillsAndTalentsInOrderToGai	0.774836
alwaysMakingProgress	0.276091
extremelyGoodCommunicator	0.248468
iDontFeelParticularlyPleasedWithTheWayIAm	0.308332
iFeelThatLifeIsVeryRewarding	0.465194
iHaveVeryWarmFeelingsTowardsAlmostEveryone	0.834434
iAmNotParticularlyOptimisticAboutTheFuture	0.817872
lifeIsGood	0.313541
iLaughALot	0.030709
iDontThinkILookAttractive	0.154677

(Intercept)

genderMale

```

genderOther
income10k - 15k ***
income120k - 150k ***
income150k - 200k ***
income15k - 20k ***
income200k above ***
income20k - 50k ***
income50k - 80k ***
income80k - 120k ***
whatIsYourHeightExpressItAsANumberInMetresM150 - 155
whatIsYourHeightExpressItAsANumberInMetresM155 - 160
whatIsYourHeightExpressItAsANumberInMetresM160 - 165
whatIsYourHeightExpressItAsANumberInMetresM165 - 170
whatIsYourHeightExpressItAsANumberInMetresM170 - 175 *
whatIsYourHeightExpressItAsANumberInMetresM175 - 180 *
whatIsYourHeightExpressItAsANumberInMetresM180 - 185 **
whatIsYourHeightExpressItAsANumberInMetresM185 - 190 *
whatIsYourHeightExpressItAsANumberInMetresM190 above
doYouFeelASenseOfPurposeAndMeaningInYourLife104Yes .
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey complement each other
howDoYouReconcileSpiritualBeliefsWithScientificOrRationalThinkiThey conflict
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsNever
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsRarely
howOftenDoYouFeelSociallyConnectedWithYourPeersAndFriendsSometimes
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have both
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have family
doYouHaveASupportSystemOfFriendsAndFamilyToTurnToWhenNeededYes, I have friends
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVAt least once a week
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVMore than three times a week
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVNo
howOftenDoYouParticipateInSocialActivitiesIncludingClubsSportsVRarely
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferNever
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferRarely .
doYouFeelComfortableEngagingInConversationsWithPeopleFromDifferSometimes *
doYouFeelASenseOfPurposeAndMeaningInYourLife105Yes
alwaysStressed ***
alwaysAccountableAndResponsibleForYourActions
alwaysCalm
myBodyIsHypermobileAndLovesToMove
alwaysHaveFun
alwaysSerious

```

```

alwaysDepressed
alwaysLoveAndCareForYourself
extremelyGoodAbilityToSense
alwaysHaveDigestiveProblems
iAmIntenselyInterestedInOtherPeople
iRarelyWakeUpFeelingRested
iFindMostThingsAmusing
iAmAlwaysCommittedAndInvolved
iDoNotThinkThatTheWorldIsAGoodPlace
iAmWellSatisfiedAboutEverythingInMyLife
iFindBeautyInSomeThings
iAlwaysHaveACheerfulEffectOnOthers
iFeelThatIAmNotEspeciallyInControlOfMyLife
iUsuallyHaveAGoodInfluenceOnEvents
iDontHaveFunWithOtherPeople
alwaysEngageInPreparingAndUsingYourSkillsAndTalentsInOrderToGai
alwaysMakingProgress
extremelyGoodCommunicator
iDontFeelParticularlyPleasedWithTheWayIAm
iFeelThatLifeIsVeryRewarding
iHaveVeryWarmFeelingsTowardsAlmostEveryone
iAmNotParticularlyOptimisticAboutTheFuture
lifeIsGood
iLaughALot
iDontThinkILookAttractive
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.21 on 432 degrees of freedom
Multiple R-squared:  0.7665,    Adjusted R-squared:  0.7303
F-statistic: 21.16 on 67 and 432 DF,  p-value: < 2.2e-16

```

```
[1] "income10k - 15k"
[2] "income120k - 150k"
[3] "income150k - 200k"
[4] "income15k - 20k"
[5] "income200k above"
[6] "income20k - 50k"
[7] "income50k - 80k"
[8] "income80k - 120k"
[9] "whatIsYourHeightExpressItAsANumberInMetresM180 - 185"
[10] "alwaysStressed"
```

Top 5 strongest predictors are : income80k - 120k income50k - 80k income200k above income20k - 50k income15k - 20k
A matrix: 5 x 4 of type dbl

	Estimate	Std. Error	t value	Pr(> t)
income80k - 120k	37.28064	1.660049	22.45755	1.419507e-74
income50k - 80k	28.70233	1.742010	16.47656	1.115904e-47
income200k above	20.03062	1.382642	14.48721	4.777672e-39
income20k - 50k	20.81995	1.587546	13.11455	2.606555e-33
income15k - 20k	15.40746	1.284859	11.99156	8.481388e-29

A full multiple linear regression model was fitted using happiness as the response variable and all other variables as predictors. Following the tutorial approach, I used summary(full_model) to inspect the coefficient table, p-values, and significance stars. Predictors with p-values smaller than 0.01 were automatically selected as variables possibly associated with happiness. The top five strongest predictors were ranked by the absolute value of their t-statistics.

Question 2 (2 Mark)

R squared from the summary table reflects that the full model doesn't fit the training dataset well; thus, you try to quantify the error between the values of the ground-truth and those of the model prediction. You want to write a function to predict 'happiness' with the given dataset and calculate the **root mean squared error (rMSE)** between the model predictions and the ground truths. Please test this function on the full model and the training dataset.

```
In [18]: # ANSWER BLOCK
regression_train <- read.csv(get_data_path_reg("regression_train.csv"))
regression_test <- read.csv(get_data_path_reg("regression_test.csv"))

full_model<- lm(happiness ~ ., data = regression_train)

rMSE <- function(prediction,truth){sqrt(mean((truth - prediction)^2))}

full_model_predictions <- predict(full_model, regression_train)
full_model_rMSE <- rMSE(full_model_predictions, regression_train$happiness)

cat("The training rMSE of the full model is:",full_model_rMSE)
```

The training rMSE of the full model is: 6.701688

The rMSE function first calculates the squared difference between the observed happiness values and the predicted values, then takes the mean and square root. The full multiple linear regression model was used to predict happiness on the training data, and the rMSE was calculated by comparing these predictions with the true happiness values.

Question 3 (2 Marks)

You find the full model complicated and try to reduce the complexity by performing [bidirectional stepwise regression](#) with [BIC](#).

Calculate the **rMSE** of this new model with the function that you implemented previously. Is there anything you find unusual? Explain your findings in 100 words.

```
In [19]: # ANSWER BLOCK
regression_train <- read.csv(get_data_path_reg("regression_train.csv"))
regression_test <- read.csv(get_data_path_reg("regression_test.csv"))

full_model<- lm(happiness ~ ., data = regression_train)

rMSE <- function(prediction,truth){sqrt(mean((truth - prediction)^2))}

full_model_predictions <- predict(full_model, regression_train)
full_model_rMSE <- rMSE(full_model_predictions, regression_train$happiness)
cat("The training rMSE of the full model is:",full_model_rMSE,"\n")
```



```
cat("Number of predictors in the full model:", length(coef(full_model)) - 1, "\n", "\n")

bic_model<-step(full_model, trace = 0, k = log(nrow(regression_train)), direction = "both")
bic_model_predictions <- predict(bic_model, regression_train)
bic_model_rMSE <- rMSE(bic_model_predictions, regression_train$happiness)
cat("The training rMSE of the bic model is:", bic_model_rMSE, "\n")
cat("Number of predictors in the bic model:", length(coef(bic_model))-1)
```

The training rMSE of the full model is: 6.701688
 Number of predictors in the full model: 67

The training rMSE of the bic model is: 7.330305
 Number of predictors in the bic model: 14

The unusual finding is that the BIC model has a higher training rMSE than the full model. The full model has a training rMSE of 6.701688, while the BIC model has a training rMSE of 7.330305. BIC does not only focus on reducing training error rMSE. It also reduces model complexity. Some predictors are removed even if they slightly improve the training fit. Therefore, the BIC model is simpler with only 14 predictors, compared to the 67 predictors of the full model, but this can also lead to a higher training rMSE.

Question 4 (2 Mark)

Although stepwise regression has reduced the model complexity significantly, the model still contains a lot of variables that we want to remove. Therefore, you are interested in lightweight linear regression models with ONLY TWO predictors. Write a script to automatically find the best lightweight model which corresponds to the model with the least **rMSE** on the training dataset. Compare the **rMSE** of the best lightweight model with the **rMSE** of the full model - `lm.fit` - that you built previously. Give an explanation for these results based on consideration of the predictors involved.

```
In [20]: # ANSWER BLOCK
regression_train <- read.csv(get_data_path_reg("regression_train.csv"))
regression_test <- read.csv(get_data_path_reg("regression_test.csv"))

full_model<- lm(happiness ~ ., data = regression_train)

rMSE <- function(prediction,truth){sqrt(mean((truth - prediction)^2))}

full_model_predictions <- predict(full_model, regression_train)
full_model_rMSE <- rMSE(full_model_predictions, regression_train$happiness)
```

```

cat("The training rMSE of the full model is:",full_model_rMSE,"\n")
cat("Number of predictors in the full model:", length(coef(full_model)) - 1,"\n","\n")

predictor_names <- names(regression_train)[names(regression_train)!="happiness"]

two_predictor_combinations <- combn(predictor_names,2)

best_two_predictors_rMSE <- Inf
best_two_predictors_model <- NULL
best_two_predictors <- NULL

for (i in 1:ncol(two_predictor_combinations)){
  two_predictors<- two_predictor_combinations[,i]
  two_predictors_formula <- as.formula(paste("happiness ~",paste(two_predictors, collapse = "+")))
  two_predictors_model <- lm(two_predictors_formula, data = regression_train)
  two_predictors_result <- predict(two_predictors_model, regression_train)
  two_predictors_rMSE<- rMSE(two_predictors_result,regression_train$happiness)

  if (two_predictors_rMSE < best_two_predictors_rMSE) {
    best_two_predictors_rMSE<-two_predictors_rMSE
    best_two_predictors_model<-two_predictors_model
    best_two_predictors<-two_predictors
  }
}

cat("The best two predictors rMSE are:",best_two_predictors_rMSE,"\n")
cat("The best two predictors are:",best_two_predictors,"\n")
summary(best_two_predictors_model)

```

The training rMSE of the full model is: 6.701688

Number of predictors in the full model: 67

The best two predictors rMSE are: 7.885411

The best two predictors are: income alwaysStressed

Call:

```
lm(formula = two_predictors_formula, data = regression_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-36.097	-5.068	0.050	5.502	24.288

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-12.9244	0.6116	-21.134	< 2e-16 ***
income10k - 15k	7.7152	1.2865	5.997	3.91e-09 ***
income120k - 150k	12.7052	1.2669	10.028	< 2e-16 ***
income150k - 200k	11.1911	1.2635	8.857	< 2e-16 ***
income15k - 20k	13.8703	1.2628	10.984	< 2e-16 ***
income200k above	20.4021	1.3996	14.577	< 2e-16 ***
income20k - 50k	22.5643	1.5631	14.435	< 2e-16 ***
income50k - 80k	28.8786	1.8088	15.966	< 2e-16 ***
income80k - 120k	36.9091	1.6307	22.634	< 2e-16 ***
alwaysStressed	-2.5286	0.2949	-8.576	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.965 on 490 degrees of freedom

Multiple R-squared: 0.6767, Adjusted R-squared: 0.6708

F-statistic: 114 on 9 and 490 DF, p-value: < 2.2e-16

ANSWER (TEXT)

The best two-predictor model selected income and alwaysStressed, with a training rMSE of 7.885411. This is higher than the full model rMSE of 6.701688, which is expected because the full model uses many more predictors and can fit the training data more closely. However, the two-predictor model is much simpler and easier to interpret. The selected predictors also make sense, because income shows a strong positive association with happiness, while alwaysStressed has a negative coefficient. Therefore, this lightweight model sacrifices some training accuracy but still captures two important sources of variation in happiness.

Question 5 (Libraries are allowed) (40 Marks)

As a Data Scientist, one of the key tasks is to build models **most appropriate/closest** to the truth; thus, modelling will not be limited to the aforementioned steps in this assignment. To simulate for a realistic modelling process, this question will be in the form of a [Kaggle competition](#) among students to find out who has the best model.

Thus, you **will be graded** by the **rmse** performance of your model, the better your model, the higher your score. Additionally, you need to describe/document your thought process in this model building process, this is akin to showing your working properly for the mathematic sections. If you don't clearly document the reasonings behind the model you use, we will have to make some deductions on your scores.

This is the [video tutorial](#) on how to join any Kaggle competition.

When you optimize your model's performance, you can use any supervised model that you know and feature selection might be a big help as well. [Check the non-exhaustive set of R functions relevant to this unit](#) for ideas for different models to try.

Note Please make sure that we can install the libraries that you use in this part, the code structure can be:

```
install.packages("some package", repos='http://cran.us.r-project.org')
```

```
library("some package")
```

Remember that if we cannot run your code, we will have to give you a deduction. Our suggestion is for you to use the standard **R version 3.6.1**

You also need to name your final model **fin.mod** so we can run a check to find out your performance. A good test for your understanding would be to set the previous **BIC model** to be the final model to check if your code works perfectly.

The models were developed and run in Kaggle Notebook. The submitted prediction files were generated using R 4.4.0, xgboost 1.7.8.1, caret 7.0-1 and e1071 1.7.16 in the Kaggle Linux environment.

```
In [21]: #get file path for local or Kaggle environment
get_data_path_reg <- function(file_name) {
  if (file.exists(file_name)) {
    return(file_name)
  }
}
```

```

kaggle_path <- file.path(
  "/kaggle/input/datasets/zksdawn4pm/regression-dataset",
  file_name
)

if (file.exists(kaggle_path)) {
  return(kaggle_path)
}

stop(paste("Cannot find file:", file_name))
}

```

In [22]: *# Build your final model here, use additional coding blocks if you need to*

```

#load packages
library(xgboost)
library(caret)

cat("R version:", R.version.string, "\n")
cat("xgboost version:", as.character(packageVersion("xgboost")), "\n")
cat("caret version:", as.character(packageVersion("caret")), "\n")

#load raw regression data
train_data <- read.csv(get_data_path_reg("regression_train.csv"))
test_data <- read.csv(get_data_path_reg("regression_test.csv"))

#separate target variable
y <- train_data$happiness
train_data$happiness <- NULL

#combine train and test data
combined <- rbind(train_data, test_data)

#create dummy variables with combined data for categorical predictors
dummies <- dummyVars(" ~ .", data = combined)
enc<- predict(dummies, newdata = combined)

#split encoded data back into train and test matrices
X<- as.matrix(enc[1:nrow(train_data), ])
X_test <- as.matrix(enc[(nrow(train_data) + 1):nrow(combined), ])

```

```
#convert matrices into xgboost data format
dtrain<- xgb.DMatrix(data = X,label = y)
dtest <- xgb.DMatrix(data = X_test)

#set xgboost parameters for model1&2
params <- list(objective="reg:squarederror",eta=0.05, max_depth= 6,subsample= 0.8, colsample_bytree= 0.8, min_chil
```

R version: R version 4.4.0 (2024-04-24)

xgboost version: 1.7.8.1

caret version: 7.0.1

model1: CV-optimal; LB 4.42270

```
In [23]: #model1: CV-optimal; LB 4.42270
#use 5-fold CV with early stopping nrounds(~208)
set.seed(1)
cv <- xgb.cv(params = params, data = dtrain, nrounds = 2000, nfold = 5,
             early_stopping_rounds = 50, verbose = 0)
reg_fin_mod1 <- xgb.train(params = params, data = dtrain, nrounds = cv$best_iteration, verbose = 0)
reg_pred1 <- predict(reg_fin_mod1, dtest)
write.csv(data.frame(RowIndex = seq_along(reg_pred1), Prediction = reg_pred1),
          "RegressionPredictLabel_model1.csv", row.names = FALSE)
```

model2: LB-optimal(800 nrounds) 4.19993

```
In [24]: #model2: LB-optimal(800 nrounds) 4.19993
set.seed(1)
reg_fin_mod2 <- xgb.train(params = params, data = dtrain, nrounds =800, verbose = 0)
reg_pred2 <- predict(reg_fin_mod2, dtest)
write.csv(data.frame(RowIndex = seq_along(reg_pred2), Prediction = reg_pred2),
          "RegressionPredictLabel_model2.csv", row.names = FALSE)
```

Model 1 used xgboost cross-validation and early stopping to choose the number of boosting rounds, while Model 2 used a fixed 800 rounds because it achieved a stronger Kaggle public score. Since both models use the same preprocessing and xgboost parameter setting, Model 2 was selected as the final regression model.

```
In [25]: # Load in the test data.
test <- read.csv(get_data_path_reg("regression_test.csv"))

# choose model2 as the final model
set.seed(1)
fin.mod <- xgb.train(params = params, data = dtrain, nrounds = 800, verbose = 0)

# If you are using any packages that perform the prediction differently, please change this line of code accordingly
pred.label <- predict(fin.mod, dtest)

# put these predicted labels in a csv file that you can use to commit to the Kaggle Leaderboard
write.csv(
  data.frame("RowIndex" = seq(1, length(pred.label)), "Prediction" = pred.label),
  "RegressionPredictLabel.csv",
  row.names = F
)
```

```
In [ ]: ## PLEASE DO NOT ALTER THIS CODE BLOCK, YOU ARE REQUIRED TO HAVE THIS CODE BLOCK IN YOUR JUPYTER NOTEBOOK SUBMISSION
## Please skip (don't run) this if you are a student
## For teaching team use only

tryCatch(
  {
    source("../supplimentary.R")
  },
  error = function(e){
    source("supplimentary.R")
  }
)

truths <- tryCatch(
  {
    read.csv("../regression_test_label.csv")
  },
  error = function(e){
    read.csv("regression_test_label.csv")
  }
)
```

```
RMSE.fin <- rmse(pred.label, truths$x)
cat(paste("RMSE is", RMSE.fin))
```

Part 2 Classification (50 Marks)

A few thousand people were questioned in a life and wellbeing survey to build a model to predict happiness of an individual, but this time we want to predict a categorical score for their mental health, rather than a continuous score. You need to build 5-class classification models to optimally predict the variable in the survey dataset called 'alwaysAnxious' based on any, or all, of the other survey question responses.

You have been provided with two datasets, `classification_train.csv` and `classification_test.csv`. Using these datasets, you hope to build a model that can predict 'alwaysAnxious' using the other variables. `classification_train.csv` comes with the ground-truth target label (i.e. 'alwaysAnxious' happiness classes) whereas `classification_test.csv` comes with independent variables (input information) only.

On the order of around 70 survey questions have been converted into predictor variables that can be used to predict 'alwaysAnxious'. We do not list all the predictor names here, but their names given in the data header can clearly be linked to the survey questions. E.g. the predictor variable 'iDontFeelParticularlyPleasedWithTheWayIAm' corresponds to the survey question 'I don't feel particularly pleased with the way I am.'

This question will also be in the form of a [Kaggle competition](#) among students to find out who has the best model.

```
In [27]: #get file path for local or Kaggle environment
get_data_path_cla <- function(file_name) {
  if (file.exists(file_name)) {
    return(file_name)
  }

  kaggle_path <- file.path(
    "/kaggle/input/datasets/zksdown4pm/classification-dataset",
    file_name
  )

  if (file.exists(kaggle_path)) {
    return(kaggle_path)
  }
}
```



```
}  
  
  stop(paste("Cannot find file:", file_name))  
}
```

```
In [28]: #load package  
#install.packages("e1071", repos = "https://cloud.r-project.org")  
library(e1071)  
cat("e1071 version:", as.character(packageVersion("e1071")), "\n")  
  
#load raw data  
train<- read.csv(get_data_path_cla("classification_train.csv"), stringsAsFactors = FALSE)  
test<- read.csv(get_data_path_cla("classification_test.csv"), stringsAsFactors = FALSE)  
  
#convert target to 5-class factor  
y <- factor(train$alwaysAnxious, levels = c(-2, -1, 0, 1, 2))  
  
#only use numeric predictors  
num_cols <- setdiff(names(train)[sapply(train, is.numeric)], "alwaysAnxious")  
x_train <- as.matrix(train[, num_cols, drop = FALSE])  
x_test <- as.matrix(test[, num_cols, drop = FALSE])  
  
#standardise train data predictors  
center_val <- colMeans(x_train)  
scale_val <- apply(x_train,2,sd)  
#avoid division by 0  
scale_val[scale_val== 0] <- 1  
  
#scale train and test data with the same mean and sd  
x_train_s <- scale(x_train,center = center_val, scale =scale_val)  
x_test_s <- scale(x_test,center = center_val, scale =scale_val)  
  
#set weights for 5-class target  
class_table <- table(y)  
class_weights <- as.numeric(sum(class_table)/(length(class_table)*class_table))  
names(class_weights) <- names(class_table)
```

e1071 version: 1.7.16

model1 0.52704

```
In [29]: #define macro-F1 function
macro_f1 <- function(truth, pred) {
  f1 <- sapply(levels(truth), function(k) {
    tp<-sum(pred == k & truth == k)
    fp<-sum(pred == k & truth != k)
    fn<-sum(pred != k & truth == k)
    #calculate precision and recall for each class
    p <- if (tp + fp == 0) 0 else tp / (tp + fp)
    r <- if (tp + fn == 0) 0 else tp / (tp + fn)
    #calculate F1 score for each class
    if (p + r == 0) 0 else 2 * p * r / (p + r)
  })
  mean(f1)}

#split data into 5-folds
set.seed(1)
fold <- integer(length(y))
#assign samples from each class evenly into 5 folds
for (lev in levels(y)) {
  idx <- sample(which(y == lev))
  fold[idx] <- rep(1:5, length.out = length(idx))}

#run 5-folds CV
scores <- numeric(5)
for (f in 1:5) {
  #training fold
  tr <- fold != f
  #validation fold
  va <- fold == f
  #train weighted RBF SVM on the training folds with chosen parameters(model1)
  m <- svm(x = x_train_s[tr, ], y = y[tr], kernel = "radial", cost=2, gamma=0.05, class.weights = class_weights, scal
    #predict validation fold and calculate macro-F1
    scores[f] <- macro_f1(y[va], predict(m, x_train_s[va, ]))
}

cat("Macro-F1 for each fold:", scores, "\n")
cat("Mean 5-fold macro-F1:", mean(scores), "\n")
```

Macro-F1 for each fold: 0.34933 0.3892346 0.409286 0.3856296 0.4206096

Mean 5-fold macro-F1: 0.390818

```
In [30]: #model1: LB 0.52704
set.seed(1)
cla_fin.mod1 <- svm(x = x_train_s, y = y, kernel = "radial", cost=2, gamma=0.05, class.weights = class_weights, scale

pred.label <- predict(cla_fin.mod1, x_test_s)
pred.label <- as.integer(as.character(pred.label))

write.csv(
  data.frame("RowIndex" = seq_along(pred.label), "Prediction" = pred.label),
  "ClassificationPredictLabel_model1.csv",
  row.names = FALSE
)
```

model2 0.59364

```
In [31]: #define macro-F1 function
macro_f1 <- function(truth, pred) {
  f1 <- sapply(levels(truth), function(k) {
    tp <- sum(pred == k & truth == k)
    fp <- sum(pred == k & truth != k)
    fn <- sum(pred != k & truth == k)
    #calculate precision and recall for each class
    p <- if (tp + fp == 0) 0 else tp / (tp + fp)
    r <- if (tp + fn == 0) 0 else tp / (tp + fn)
    #calculate F1 score for each class
    if (p + r == 0) 0 else 2 * p * r / (p + r)
  })
  mean(f1)}

#split data into 5-folds
set.seed(1)
fold <- integer(length(y))
#assign samples from each class evenly into 5 folds
for (lev in levels(y)) {
  idx <- sample(which(y == lev))
  fold[idx] <- rep(1:5, length.out = length(idx))}
```

```

#run 5-folds CV
scores <- numeric(5)
for (f in 1:5) {
  #training fold
  tr <- fold != f
  #validation fold
  va <- fold == f
  #train weighted RBF SVM on the training folds with chosen parameters(model2)
  m <- svm(x = x_train_s[tr, ], y = y[tr], kernel = "radial", cost=1.75, gamma=0.045, class.weights = class_weights,
    #predict validation fold and calculate macro-F1
    scores[f] <- macro_f1(y[va], predict(m, x_train_s[va, ]))
}

cat("Macro-F1 for each fold:", scores, "\n")
cat("Mean 5-fold macro-F1:", mean(scores), "\n")

```

Macro-F1 for each fold: 0.3375165 0.39271 0.4111579 0.4031403 0.3942099

Mean 5-fold macro-F1: 0.3877469

```

In [32]: #model2: kaggle public score: 0.59364
set.seed(2)
cla_fin.mod2 <- svm(x = x_train_s, y = y, kernel = "radial", cost=1.75, gamma=0.045, class.weights = class_weights, s

pred.label <- predict(cla_fin.mod2, x_test_s)
pred.label <- as.integer(as.character(pred.label))

write.csv(
  data.frame("RowIndex" = seq_along(pred.label), "Prediction" = pred.label),
  "ClassificationPredictLabel_model2.csv",
  row.names = FALSE
)

```

Model 1 and Model 2 had very similar local 5-fold CV performance. Model 1 achieved a CV macro-F1 of about 0.3908, while Model 2 achieved about 0.3877, so the difference was very small. Therefore, local CV did not provide strong evidence that Model 2 was worse than Model 1.

The Kaggle public leaderboard score for Model 2 was much higher. I do not assume that this public score will exactly represent the final private score, because the public leaderboard only evaluates part of the test set and may have a different class distribution from the CV

folds. However, since Model 2 had comparable local CV performance and the strongest public leaderboard result, I selected it as the final weighted RBF SVM model.

```
In [33]: # Load in the train and test classification data
train<- read.csv(get_data_path_cla("classification_train.csv"), stringsAsFactors = FALSE)
test<- read.csv(get_data_path_cla("classification_test.csv"), stringsAsFactors = FALSE)

# Build your final model here, use additional coding blocks if you need to
#final selected model: Model 2
set.seed(2)
fin.mod <- svm(x = x_train_s, y = y, kernel = "radial", cost=1.75, gamma=0.045, class.weights = class_weights, scale

# If you are using any packages that perform the prediction differently, please change this line of code accordingly
# use processed test matrix
pred.label <- predict(fin.mod, x_test_s)
# convert factor predictors back to numeric labels
pred.label <- as.integer(as.character(pred.label))

# put these predicted labels in a csv file that you can use to commit to the Kaggle Leaderboard
write.csv(
  data.frame("RowIndex" = seq(1, length(pred.label)), "Prediction" = pred.label),
  "ClassificationPredictLabel.csv",
  row.names = F
)
```

```
In [ ]: ## PLEASE DO NOT ALTER THIS CODE BLOCK, YOU ARE REQUIRED TO HAVE THIS CODE BLOCK IN YOUR JUPYTER NOTEBOOK SUBMISSION
## Please skip (don't run) this if you are a student
## For teaching team use only

truths <- tryCatch(
  {
    read.csv("../classification_test_label.csv")
  },
  error = function(e){
    read.csv("classification_test_label.csv")
  }
)
```

```
f1_score <- F1_Score(truths$x, pred.label)
cat(paste("f1_score is", f1_score))
```